

Epistemic Exams

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5TH SOUTHEAST THEORY FESTIVAL

Introduction

Introduction

An **exam** measures what you know.

Introduction

An **exam** measures what you know.

An **epistemic exam** measures what you know *and* what you know you know.

Example

Student 1 gets 6/10 correct answers on an exam

Result. C C C C C C W W W W

and after the exam reports “I’m very confident my first 6 answers are correct. Moreover, though these are my best guesses, I’m very confident my last 4 answers are wrong.”

Beliefs. C C C C C C W W W W

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Student 2 gets 8/10 correct answers on an exam

Result. C C C C C C C C W W

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Student 2 gets 8/10 correct answers on an exam

Result. C C C C C C C C W W

and after the exam reports *"I'm very confident my last 8 answers are correct. Moreover, though these are my best guesses, I'm very confident my first 2 answers are wrong."*

Beliefs. W W C C C C C C C C

Example

Student 1 : **Result.** C C C C C C W W W W
Beliefs. C C C C C C W W W W

Student 2 : **Result.** C C C C C C C C W W
Beliefs. W W C C C C C C C C

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Student 1 : **Result.** C C C C C C W W W W
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- Student 2 **knows** more than Student 1.

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- Student 2 **knows** more than Student 1.
 - That is, Student 2 will get a higher fraction of her answers correct than Student 1, given similar questions
- Student 1 **knows what she knows** more than Student 2.
 - That is, Student 1 can perfectly identify which of her answers will be correct and which will be wrong, while Student 2 cannot

Example

How should we score each of these students?

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How should we score each of these students?

That's what this paper is about

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Food for Thought and Preview

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Food for Thought and Preview

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- Would you prefer to hire Student 1 or Student 2?
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Food for Thought and Preview

- Would you prefer to hire Student 1 or Student 2?
- It depends on the context
 - Sometimes it's better to have the student who knows more, even if they know what they know less
 - Other times it's better to have the student who knows what they know more, even if they know less
- The desired scoring rule will be a function of the characteristics of the situation you face

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- ① Introduce the concept of an epistemic exam

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For AI, the purpose is twofold:

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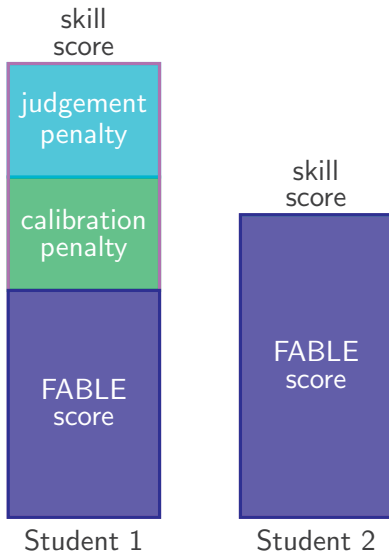
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- 2 To help train AIs to behave closer to how we would like them to behave

In particular, current AIs know a lot, but don't always know what they know, and as a result are **unreliable**

Preview





Roadmap

Introduction
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Scoring Rules (SR)
OOOO

Incentive-Compatible SR
○○○○○○○○○○○○

Utilitarian SR
○○○○○○○○○○

Roadmap

1 Scoring Rules

Introduction
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Roadmap

- 1 Scoring Rules
- 2 Incentive-Compatible Scoring Rules

Roadmap

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Roadmap

- 1 Scoring Rules
- 2 Incentive-Compatible Scoring Rules
- 3 Utilitarian Scoring Rules
- 4 FABLE

Scoring Rules

Forecasting

Test-Taking

Incentive-Compatible Scoring Rules

Forecasting

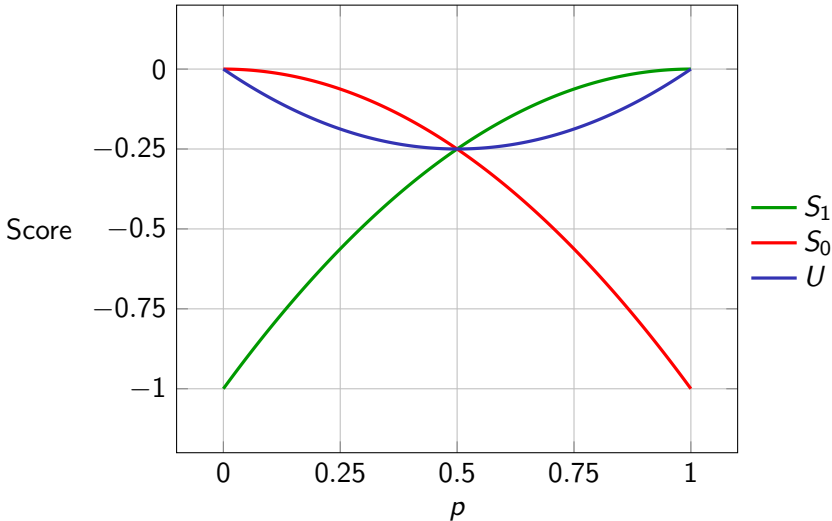
Quadratic Scoring Rule

For example, the *quadratic scoring rule* is defined by

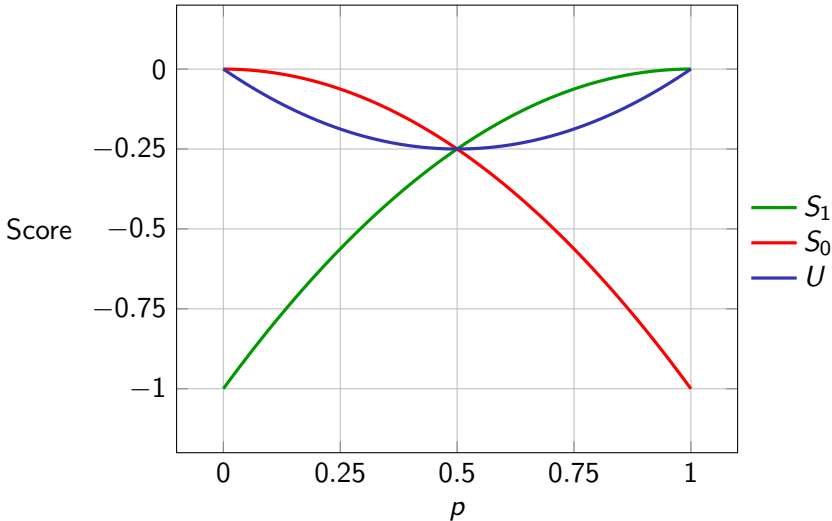
$$S_0(p) = -p^2 \quad \text{and} \quad S_1(p) = -(1-p)^2$$

and is a strictly proper scoring rule.

Quadratic Scoring Rule



Quadratic Scoring Rule

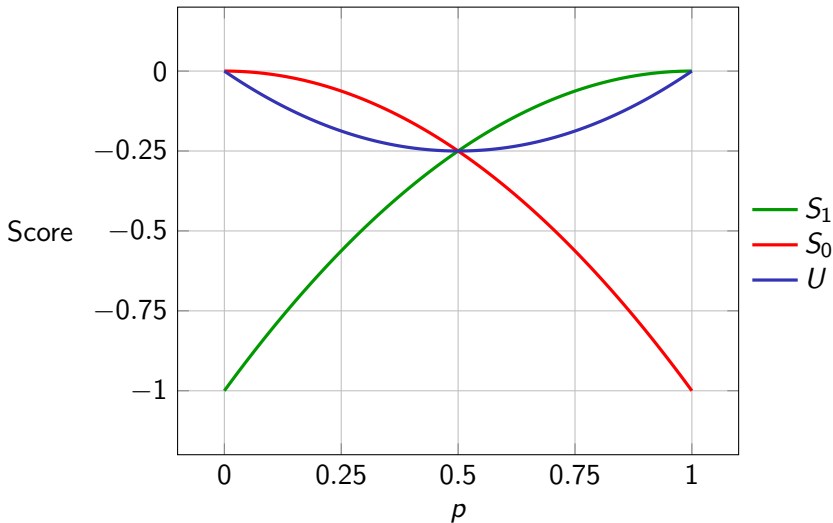


Test-Taking

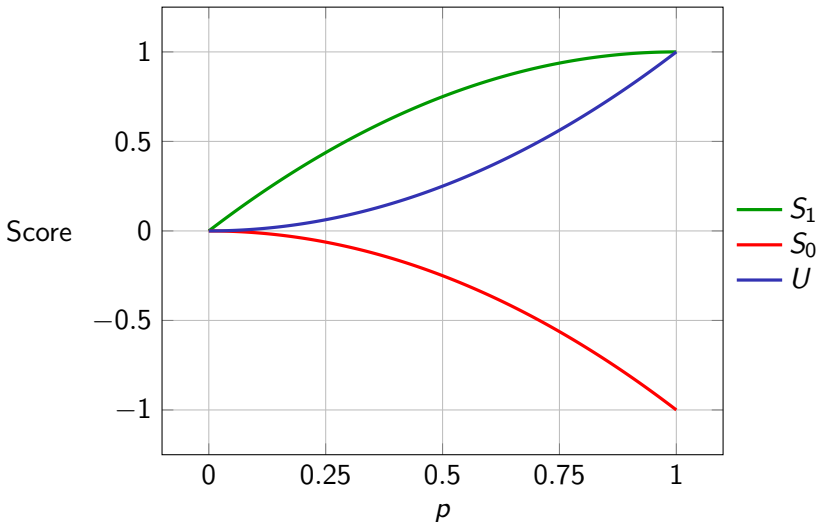
Characterization of Proper Epistemic Exam Scoring Rules

Theorem 2. A scoring rule (S_0, S_1) is a (strictly) proper epistemic exam scoring rule if and only if

Quadratic Scoring Rule



Quadratic Epistemic Exam Scoring Rule



Zooming Out

For any desired expected payoff for a truthful agent U , then there are infinitely many scoring rules (S_0, S_1) that generate it (e.g., $S_1 = S_0 = U$)

Forecasting

Proposition 1

Proposition 1. In a forecasting environment, every utilitarian scoring rule is a proper scoring rule.

Quick Aside

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I'd be curious afterwards if you have any references or thoughts on this

Test-Taking

Test-Taking Environment with User

The test-taking environment is again the forecasting environment plus a set X of possible answers the user can receive from the agent:

- X is the set of possible answers the agent can give
- A is the set of feasible actions the user can take upon receiving answer x and confidence p

Takeaway

In economics, we are usually quite concerned with incentive compatibility

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With AI, it's not obvious that we need their scores to be incentive-compatible

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With AI, it's not obvious that we need their scores to be incentive-compatible

What we do want is that the score measures their usefulness (the payoff to society of using that AI)

FABLE

Deriving FABLE

We are now going to derive FABLE

The User's (Society's) Payoffs

Let's define the user's payoffs

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First, the two normalized payoffs

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First, the two normalized payoffs

- 0 : the status quo payoff of not having an AI agent to help the user, i.e., the payoff of disregarding any answer

The 5 Parameters

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- F : cost of failure

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 - the cost of implementing an incorrect answer (relative to the benefit of implementing a correct answer)
 - $F = \frac{1-\alpha}{\alpha}$ where α is the failure rate at which the user is just indifferent between using the AI and not

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- E : cost of evaluation
 - the cost of evaluating an answer (relative to the benefit of implementing a correct answer)
 - $E = \frac{1}{n}$ where n is the number of answers at which the user would be just indifferent between checking n answers to get one correct answer and not bothering

The 5 Parameters (continued)

- $\alpha : \text{type} \mid \text{error}$

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 - the rate of false positives associated with the evaluation technology (evaluation says answer is correct when it's incorrect)
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 - the rate of false negatives associated with the evaluation technology (evaluation says answer is incorrect when it's correct)
- L : luck
 - the confidence level associated with random guessing ($1 / \text{the number of multiple choice answers}$)

The 5 Parameters

And if we reshuffle those 5 parameters...

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- F : cost of failure
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- E : cost of evaluation

The 5 Parameters for the Examples Ahead

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For the examples ahead, I will use:

- $F = 9$ (failure is 9x worse than success)
- $\alpha = 0.05$
- $\beta = 0.2$

Deriving FABLE

Let's now write the payoffs/scores assuming the user always disregards, evaluates, and implements, respectively

Disregard

Disregard

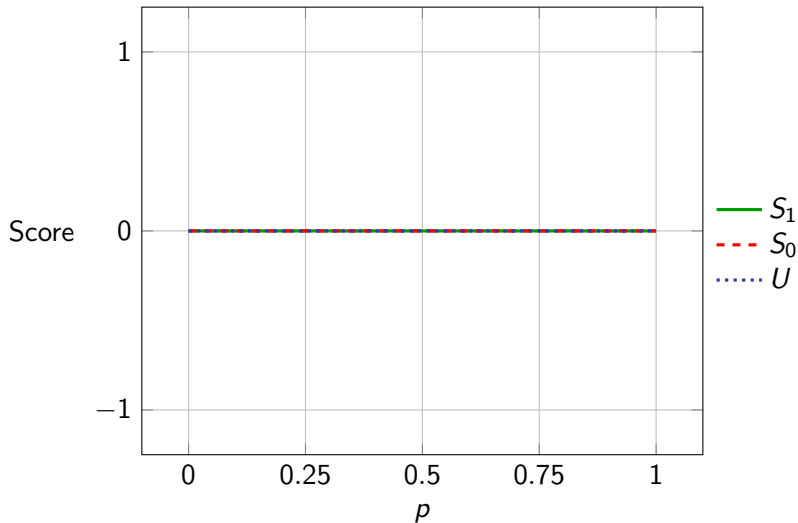
$$S_1(p) = 0$$

Disregard

$$S_1(p) = 0$$

$$S_0(p) = 0$$

Disregard

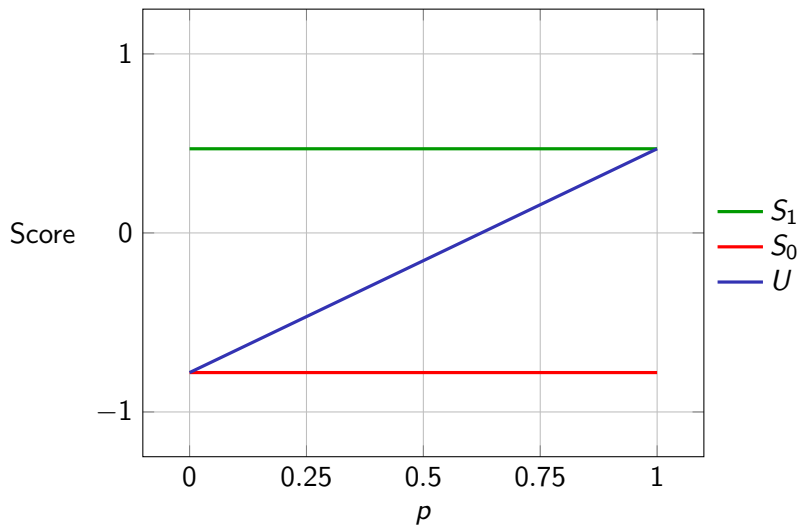


Evaluate

$$\begin{aligned} S_1(p) &= -E + (1 - \beta) \times 1 + \beta \times 0 \\ &= 1 - \beta - E \end{aligned}$$

$$\begin{aligned} S_0(p) &= -E + (1 - \alpha) \times 0 + \alpha \times (-F) \\ &= -\alpha F - E \end{aligned}$$

Evaluate



Implement

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Implement

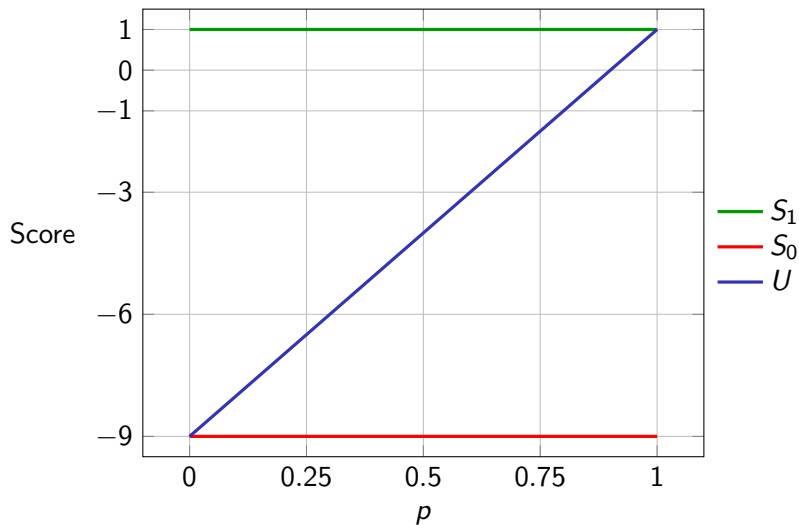
$$S_1 = 1$$

Implement

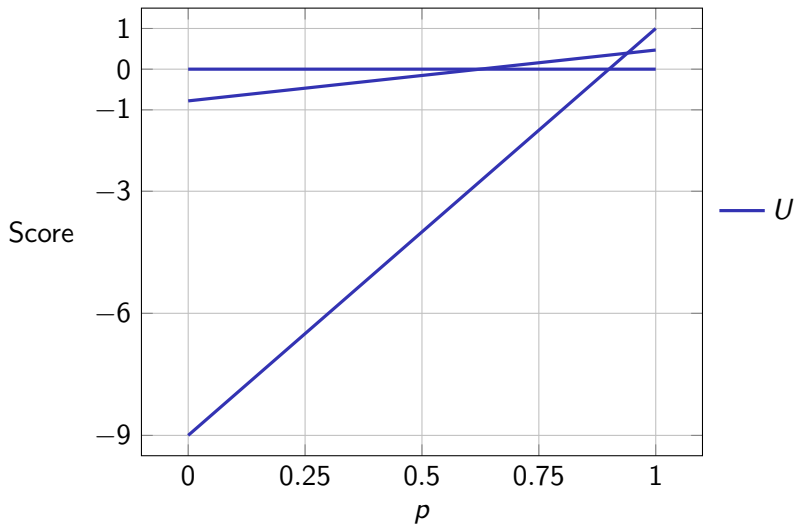
$$S_1 = 1$$

$$S_0 = -F$$

Implement



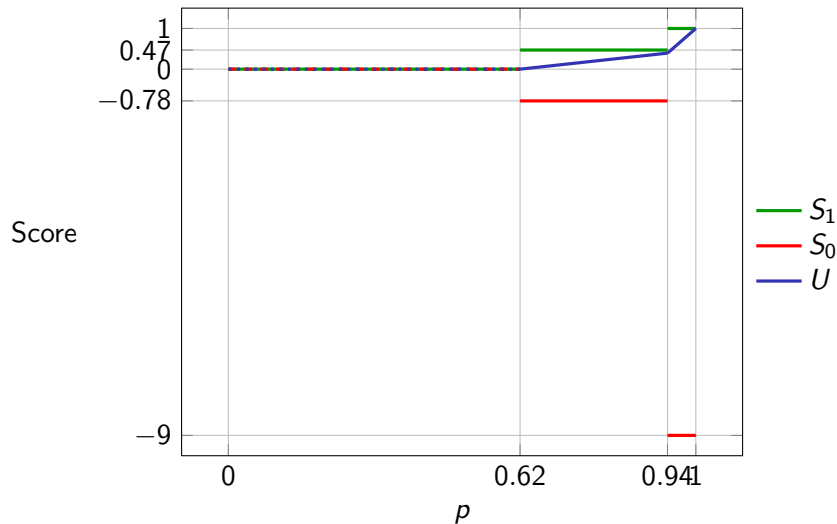
All Payoffs Together



Taking the Max



Adding S_0 and S_1 Back In



Incentives

Notice that U is weakly convex and weakly increasing

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Hence, this scoring rule gives the agent:

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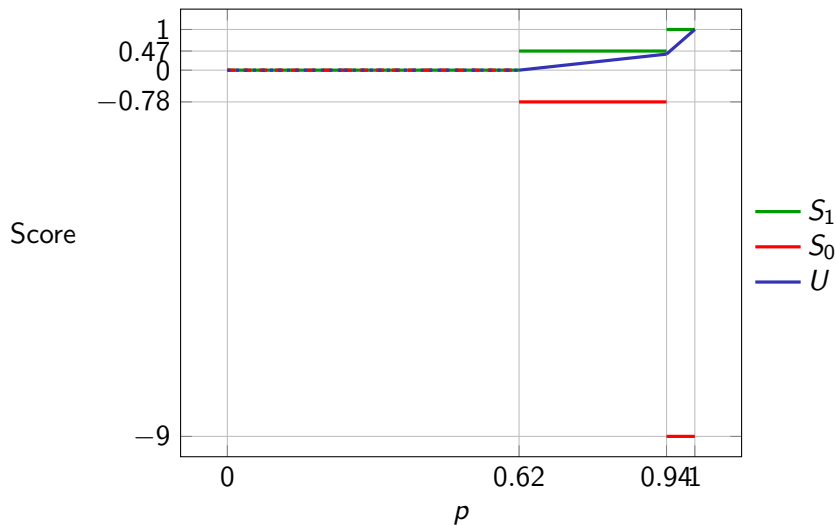
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Since we didn't use the L parameter for this, I will call this the FABLE scoring rule

FABE (9, 0.05, 0.2, 0.33)

Confidence	Outcome	Score
None [0, 0.62)	×	0
	✓	0
Intermediate [0.62, 0.94]	×	-0.78
	✓	0.47
High (0.94, 1]	×	-9
	✓	1

FABE (9, 0.05, 0.2, 0.33)



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FABE
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From FABLE to FABLE

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Moreover, I argue that this will be a closer approximation to the user's actual payoff than FABLE

From FABE to FABLE

Let U^{FABE} denote the expected payoffs of a truthful agent under FABE, which is what we derived in the previous section

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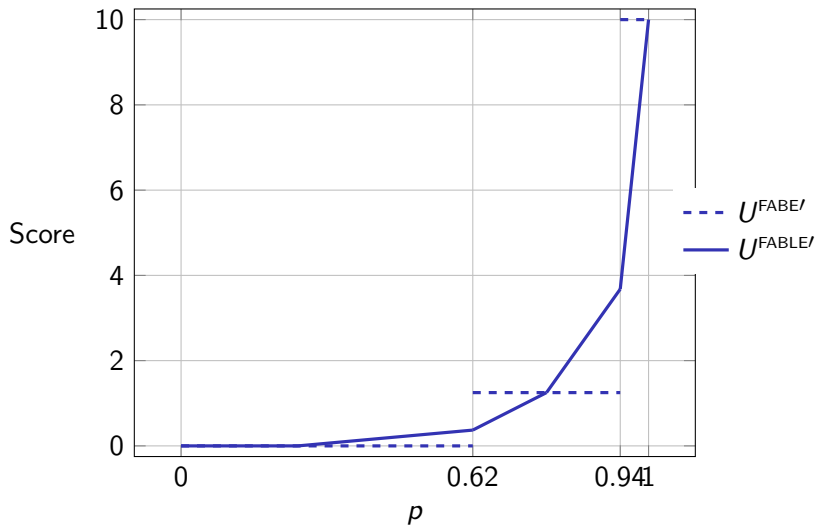
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- $U^{\text{FABLE}}(1) = U^{\text{FABE}}(1)$ and $U^{\text{FABLE}'}(1) = U^{\text{FABE}'}(1)$

It turns out this pins down U^{FABLE} uniquely

$U^{\text{FABE}'}$ and $U^{\text{FABLE}'}$



Introduction
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FABE is a simple skeleton of the payoffs for the end user, supposing the user has just three available actions: disregard, evaluate, and implement

Of course, in reality, the user has many more actions, including

- evaluating the answer with different levels of effort
- using an incorrect answer as helpful inspiration

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The FABLE skeleton is optimized to be a good approximation of U at

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- the left endpoint for disregard

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- the midpoint for evaluate

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- the right endpoint for implement

Everywhere else, it is a *lower bound* on U

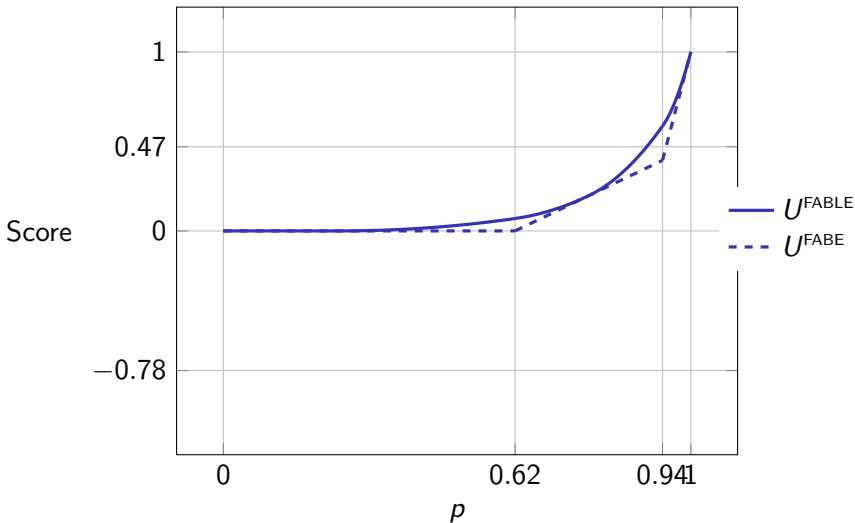
Why Do This?

The FABLE skeleton is optimized to be a good approximation of U at

- the left endpoint for disregard
- the midpoint for evaluate
- the right endpoint for implement

Everywhere else, it is a *lower bound* on U

FABLE adds the “meat” to FABLE’s “bones,” smoothing out U in between these points to capture the other actions an individual might take that are unnecessary to model explicitly

U^{FABE} and U^{FABLE} 

From U^{FABLE} to (S_0, S_1)

We've pinned down U^{FABLE} , but haven't yet pinned down (S_0, S_1)

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Recall that there are infinitely many (S_0, S_1) that could generate U^{FABLE}

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By Theorem 2, since U^{FABLE} is strictly convex and strictly increasing, there exists a unique (S_0, S_1) which is strictly incentive compatible

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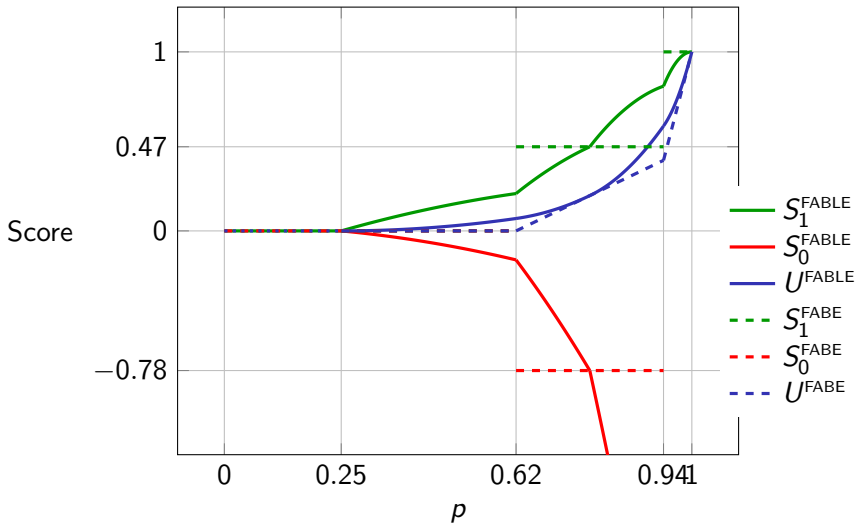
By Theorem 2, since U^{FABLE} is strictly convex and strictly increasing, there exists a unique (S_0, S_1) which is strictly incentive compatible

By Proposition 2, the scoring rule must be incentive compatible if it is to reflect the payoffs of society

FABE and FABLE



FABE and FABLE



Online Demo

Online Demo

Introduction
○○○○○○○○

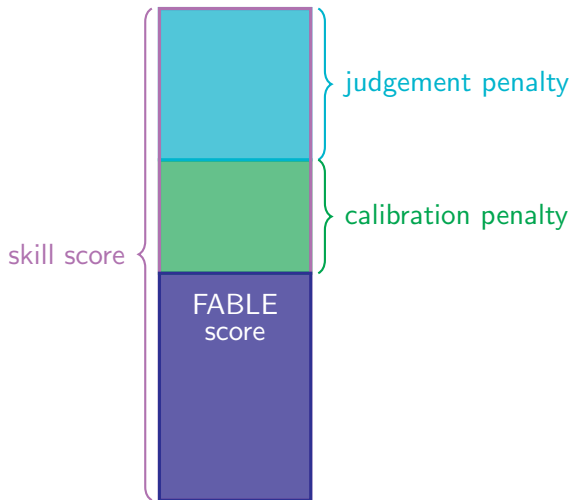
Scoring Rules (SR)
○○○○

Incentive-Compatible SR
○○○○○○○○○○

Utilitarian SR
○○○○○○○○

FABLE
○○●○○○○

Skill, Judgement, and Calibration



Example

